Succinct and Fast Tiny Pointer Hash Tables

Fast hash tables, without the memory tax

Xilin Tang
Cornell University

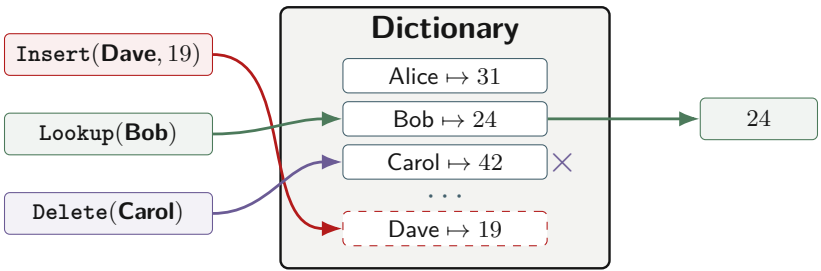
Yuqi Mai
Cornell University

William Kuszmaul
Carnegie Mellon University

Alex Conway
Cornell Tech

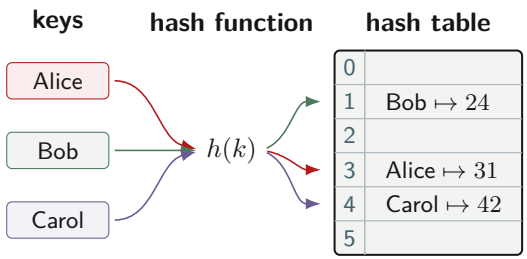
A **dictionary** stores **key–value pairs** and supports updates and lookups.

Abstractly: entries are $(k, v) \in K \times V$, and lookup is a partial function $f : K \to V \cup \{\perp\}$.



What is a hash table?

A **hash table** is a construction for dictionaries: it uses a **hash function** to help assign each key–value pair to an array slot.



Hashing dates to 1953

Expected $O(1)$ operations

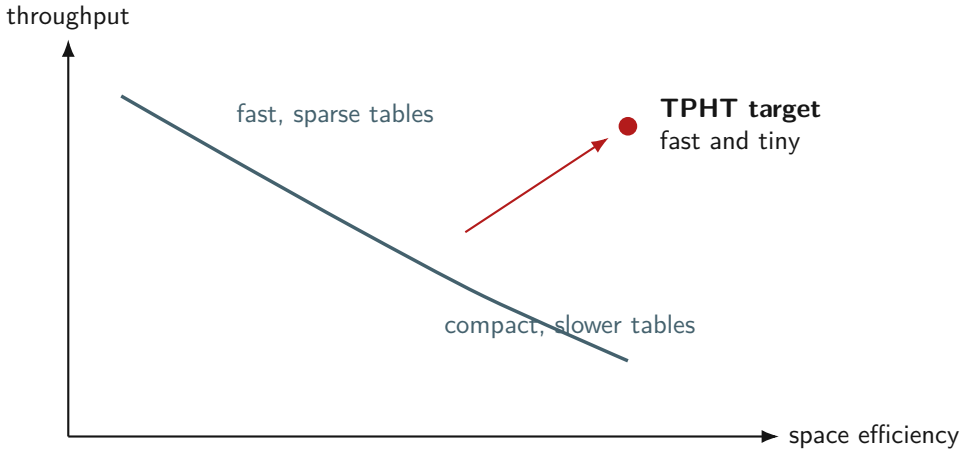
Linear probing, chaining, ...

Decades of tuning for { **speed** **space**

Practical hash tables can still be even

{ **faster** **smaller**

The Tradeoff We Want to Break

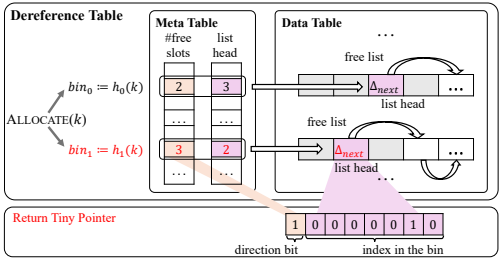


Collision resolution pays in empty slots, metadata, or full pointers.

1. Tiny pointers	2. Key quotienting	3. Two layouts
Replace 64-bit pointers with 8-bit references through a specialized dereference table.	Store only the key bits that are not already implied by the item's location.	Use the same primitives to make either the smallest table or the fastest compact table.

Chained-TPHT 105.4% space efficiency

Flattened-TPHT up to 89.3% higher throughput

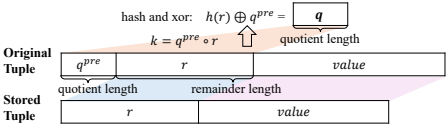


A pointer becomes one byte

- store objects in a fixed-size **dereference table**
- allocate using the **power of two choices**
- encode choice bit + in-bin offset as an 8-bit pointer
- dereference is arithmetic once the owner ID is known

98.2% supported load factor for insertion-only workloads

95% supported load factor with alternating inserts/deletes



The location already stores information

If a key belongs to quotient group q, the high-order information represented by q does not need to be stored again.

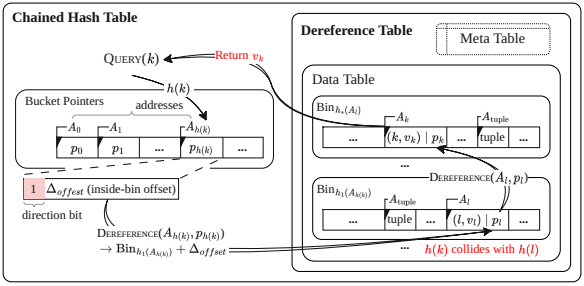
The systems problem

Filters can quotient fingerprints. Hash tables must recover the real key.

TPHT's move

Use a one-round Feistel permutation so the key can be quotiented and still decoded.

$\log n$ bits saved per key, in the useful regime



Make chaining small enough to be modern

- head array stores 1-byte tiny pointers
- nodes store next pointer, quotiented key remainder, value
- quotient groups are exactly chains
- quotienting pays back the pointer overhead

105.4% space efficiency
38.6% mean memory reduction vs. strong baselines

Why Chaining Becomes Good Again

Conventional chaining

- 8-byte pointers dominate metadata
- head arrays become expensive
- pointer chasing hurts locality

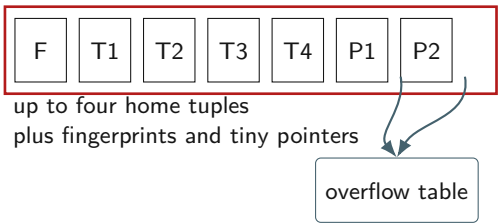
Chained-TPHT

- 1-byte head pointers make many buckets cheap
- compact head array is more cache-resident
- quotienting removes enough key bits to offset links

Chained-TPHT turns the oldest collision-resolution idea into a practical succinct hash table with expected $O(1)$ operations.

Design 2: Flattened-TPHT

64-byte home block



up to four home tuples
plus fingerprints and tiny pointers

Most operations inspect one cache line; overflow tuples cost one extra miss rather than a chain walk.

Spend a little space where latency sees it

- home blocks fit in one cache line
- overflow references are tiny pointers
- SIMD fingerprints prune comparisons
- queries average about 1.25 cache misses

83.4% space efficiency
310.4M average run-phase ops/s

	Chained-TPHT	**Flattened-TPHT**
Main goal	smallest practical table	fastest compact table
Layout	quotient groups as chains	cache-line home blocks
Critical primitive	tiny pointers everywhere	tiny pointers for overflow
Key trick	quotienting pays for links	home tuples avoid indirection
Best headline	105.4% space efficiency	up to 89.3% faster than baselines

The contribution is not only a point design: it is a reusable recipe for compact, update-friendly tables.

Practicality Is Part of the Result

Operations

64-bit keys and values, inserts, queries, updates, deletes.

Resizing

Online resizing avoids global pauses; Chained-TPHT staggers memory growth.

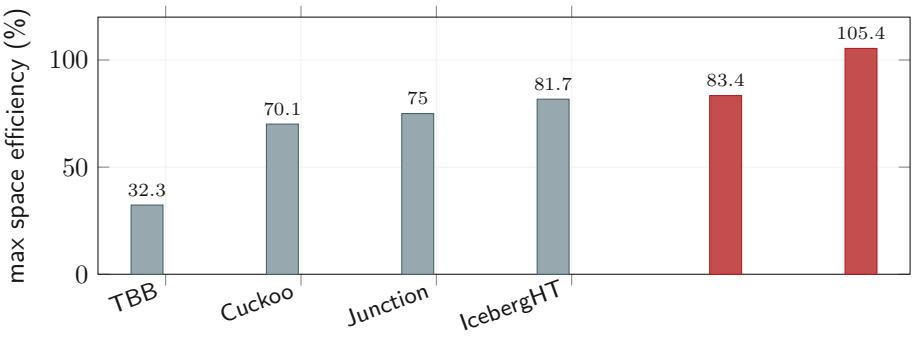
Concurrency

Near-linear scaling to 16 threads in the evaluation.

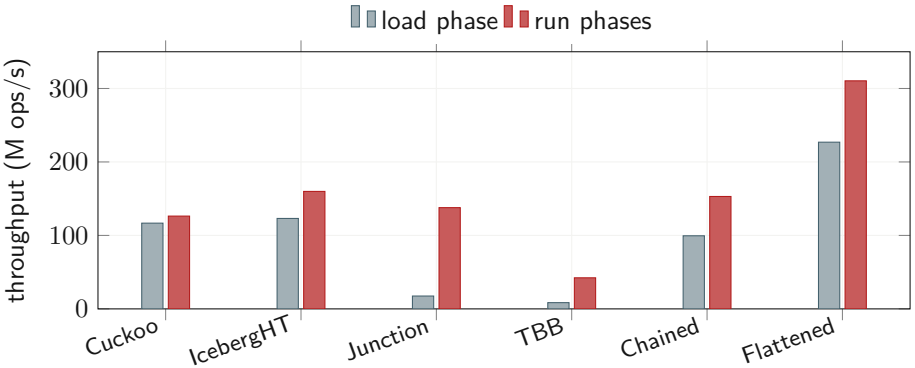
541 ns Flattened-TPHT 99p insert latency during resizing

778 ns Chained-TPHT 99p insert latency during resizing

0.925 Flattened-TPHT thread scaling factor



Chained-TPHT goes beyond the raw key-value payload size; Flattened-TPHT remains compact while targeting latency.



On YCSB with 16 threads, Flattened-TPHT has the highest average load and run throughput.

What to Remember

1

Tiny pointers make pointer-rich structures worth revisiting.

2

Quotienting turns location into saved key bits without losing recoverability.

3

The same primitives produce both a succinct table and a latency-first table.

TPHT shows that ideas once treated as “too theoretical” can redraw a very practical Pareto frontier.

Thank you.

Questions?

Xilin Tang Cornell University

Artifact: `github.com/Xilinion/TinyPtr`